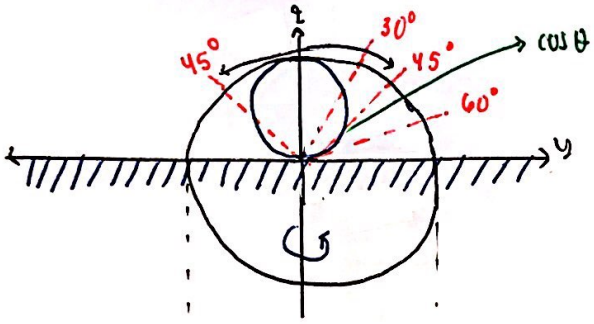
NAMA : Adira Salsabila
NIM : 181331003
KELAS : 2A - D3 Teknik Telekomunikasi
TTD :
MATA KULIAH : ANTENA DAN PROPAGASI

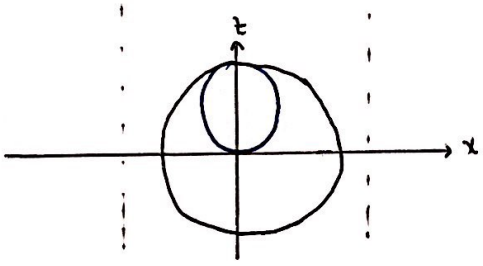
① Antena Unidirectional

$g(\theta, \phi) = \cos^2\theta$, $0 \le \theta \le \frac{\pi}{2}$ dan $0 \le \phi \le 2\pi$

ⓐ



ⓑ



ⓒ HPBW-1 = $\cos^{-1}\left(\left(\frac{1}{\sqrt{2}}\right)^{\frac{1}{2}}\right) = 32,765°$

Sehingga, HPBW-1 = 2 (θ)
= 2 (32,765°)

HPBW-1 = 65,53°

ⓓ HPBW-2 → HPBW-1 = HPBW-2
65,53° = HPBW-2
Sehingga, **HPBW-2 = 65,53°**

ⓔ Direktivitas = $\frac{41253}{(\text{HPBW-1})(\text{HPBW-2})} = \frac{41253}{(65,53)^2} =$ **9,606**

Sehingga, D = 10 log (9,606)

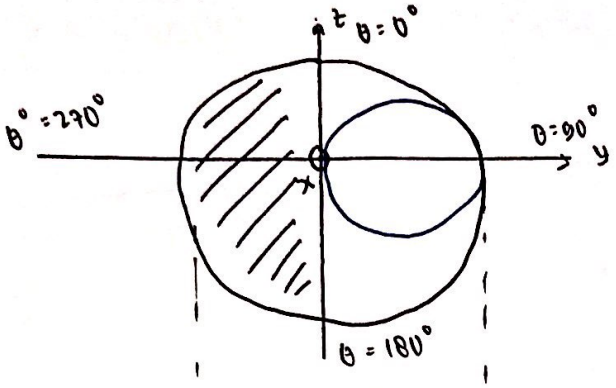
D = 9,825 dB. ≈ **10 dB**

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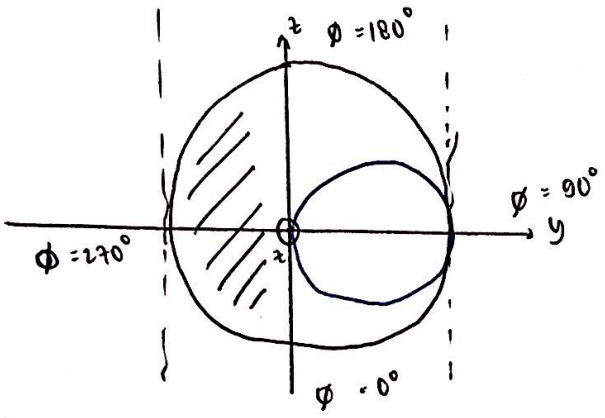
(2) Antena Unidireksional

$g(\theta, \phi) = \sin^3\theta \sin\phi$, $0 \le \theta \le \pi$ dan $0 \le \phi \le \pi$.

(a)



(b)



(c) HPBW-1 → $\sin^{-1}\left(\left(\frac{1}{\sqrt{2}}\right)^{\frac{1}{3}}\right) = 62,986°$

Sehingga, (HPBW-1) = $2(90° - \theta)$

$= 2(90° - 62,986°)$

$= 2(27,014°)$

$= \boxed{54,028°}$

(d) HPBW-2 → $\sin^{-1}\left(\left(\frac{1}{\sqrt{2}}\right)^{\frac{1}{3}}\right) = 45°$

Sehingga, (HPBW-2) = $2(90° - \theta)$

$= 2(90° - 45°)$

$= 2(45°)$

$= \boxed{90°}$

(e) direktivitas → $D = \frac{41253}{(HPBW\text{-}1)(HPBW\text{-}2)} = \frac{41253}{(54,028°)(90°)} = \frac{41253}{(4862,52°)} = \boxed{8,4838}$

$D = 10 \log(8,4838)$

$= \boxed{9,2859}\ dB \approx \boxed{9\ dB}$

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